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Editors, npj Quantum Information

Dear Editors,

We submit the enclosed manuscript, "Certified Adversarial Risk Bounds for Quantum Error-Correcting Memories: Injection-Budget Certification with Simultaneous Intervals and Reproducible Three-Code Bivariate-Bicycle Benchmarks," for consideration in npj Quantum Information.

The manuscript reframes quantum memory evaluation as a certification problem: for a fixed implementation and a bounded-weight fault-injection adversary, the operational risk $\lambda_{\mathcal{M}}(W)$ is estimated on device and reported as simultaneous intervals $[L_W, U_W]$ at joint $\geq 95\%$ confidence, with upper bounds stated honestly as 1 wherever the weight region is not exhausted. Four contributions we believe clear the journal's bar:

1. **A certification framework with simultaneous intervals and a machine-checkable risk certificate.** Intervals are Bonferroni-simultaneous across all 676 statistical cells with exact Clopper–Pearson bounds at zero counts ($2.996/N$, not the anti-conservative Wilson value), and uniform-mixture confidence sequences govern the adaptive ladder. Where enumeration is complete — the $t = 2$ radius and the noiseless injection boundary — the certified value is exact, not a bound; where it is not, the upper bound is stated as 1 with the gap printed. The per-system lower/upper envelope is shipped as a single machine-checkable artifact (`risk_certificate_v24.csv`) whose every entry regenerates from the archived counts. Estimands were hash-pinned before submission; where a conditioning estimand was instead adopted retrospectively (campaign 1), the manuscript says so explicitly in a dated chronology ledger. To our knowledge this is the first QEC benchmark whose headline numbers are simultaneous intervals with an explicit threat model.
2. **An exhaustive, independently checkable code certificate.** BB[[72,12,6]] is certified $d = 6$ exactly (meet-in-the-middle through weight 5; exhaustive weight-6 enumeration exhibiting $84 + 84$ nontrivial logicals). The archive publishes all witnesses and ships a self-contained 334-line checker that re-derives every obligation from the pinned matrices, so the distance claim is third-party verifiable in linear time per witness.
3. **A reproducible three-code benchmark at pinned scales.** Certification-first, bit-identical paired comparisons across BB72/BB144/BB288: BP+OSD lies 67.8× to $>$ 134,000× below graphified matching baselines (joint $\geq 95\%$); a seven-decoder grid on 70,000 paired instances shows BP augmentation closes the gap to BP+OSD while bare symmetry matching is non-viable; a five-decoder re-analysis of the released Kunlun $[[18, 4, 4]]$ detector model confirms the graphification penalty on hardware matrices; and a prior-aligned repeated-cycle ladder measures block-level distance suppression ($\Lambda$ per $\Delta d$=2, honestly delimited against per-cycle conventions).
4. **Hash-pinned hardware injection campaigns.** On ibm_marrakesh and ibm_fez, a pinned injection-budget protocol confirmed the distance cliff (92–96% collapse) and the device-aware composed bound on every arm, replicating across calibration cycles, devices, and distance 7; calibration-informed placement lowered logical error in four of four configurations ($p = 0.042$). Two v23-era defects found in review — a reversed extraction convention in one circuit-level

campaign and a population-average estimand misapplied to pinned masks — are disclosed in the manuscript, corrected with protocol-invariance tests and a fixed-mask reanalysis, and the affected claims are withdrawn or replaced. A second-round review found two further defects — an endpoint classifier keyed to the observed rather than the true residual syndrome, and "pre-registered" phrasing applied to a retrospectively adopted estimand — which are likewise corrected (regression-tested classifier, regenerated tables, and a full chronology ledger) and disclosed.

The package is provenance-anchored: the simulation and analysis figures and tables regenerate from the accompanying archive (SHA-256 manifests, deterministic seeds, recorded job IDs and QPY circuits for all hardware runs, an offline master reproduction script, and pinned dependency versions). Hardware outcomes themselves are experimental records and are re-attainable only by re-running the pinned protocols on device, which the archive fully specifies; the corrected circuit-level grid regenerates rate by rate under a checkpointed protocol, and the manuscript states precisely which rates are complete at submission.

[AUTHORS: one paragraph situating the work in your group's research line; disclose the prior TQE submission history if required by policy; confirm the Nature Quantum Computing submission is fully closed and state so explicitly.]

The manuscript is not under consideration elsewhere. [AUTHORS: suggested reviewers / opposed reviewers per journal policy.]

Sincerely,

[AUTHORS]